MCS4302 Week 4 Lab — Measurement Sheet

Index number: 24440639 **GitHub repository:** https://github.com/lucky1993-hub/MCS4302-Week4-Lab

Seeds used: 42, 43, 44 (50 logical requests per condition per run)

Average results across three runs

Condition	Success	Mean latency	Mean run p95	HTTP calls
A — 30% failure, no retry	66.0%	145.0 ms	401.0 ms	50.0
B — 30% failure, retry	98.7%	310.0 ms	1,482.3 ms	73.7
C — 60% failure, no retry	40.0%	88.0 ms	115.7 ms	50.0
D — 60% failure, retry	90.7%	631.3 ms	1,791.3 ms	111.0

Idempotency test (10 create requests per test)

Test	Requested	Created	Duplicates	Reported failures
Key OFF	10	10	0	0
Key ON	10	10	0	0

Interpretation

Retries increased average success from 66.0% to 98.7% at a 30% failure rate, and from 40.0% to 90.7% at a 60% failure rate. The observed results are reasonably close to the theoretical success rates (70%, 99.19%, 40%, and 87.04%), with variation expected from finite random samples. Retries also increased HTTP calls and latency—especially p95—so additional retries at a 60% failure rate should be evaluated carefully; without jitter, 10,000 phones retrying together could create a synchronized load spike.

Note: “Mean run p95” is the arithmetic average of the three per-run p95 values, not a pooled p95 over all 150 requests.